

TEAM



ALEJANDRO BENJAMÍN VARGAS GALLARDO
Capitán. - Programador y Diseñador Elec.

Primarily responsible for programming the line follower, green and red tape detection, and communication between microcontrollers, in addition to PCB fabrication.



HILEL OMAR ARIAS CASTRO
Cocapitán. - Creación de modelos IA.

He performed by creating and training our machine vision models from scratch for silver tape detection and victim recognition.



IVANNA ZOÉ SALAZAR ALMAGUER
Diseñadora Mecánica

Their job consisted of making the necessary parts for our robot, made with the Fusion 360 software to later be 3D printed.

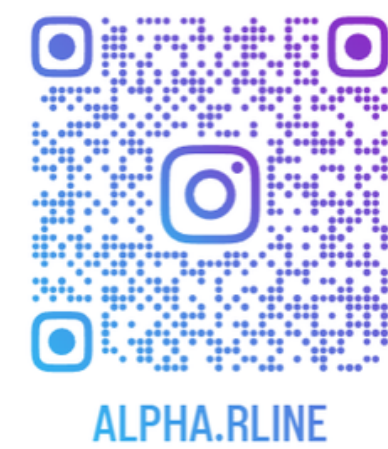


ÁNGEL ISAAC DEL RÍO GALVÁN
Programador.

Their focus was on programming motors, performing the sequences of turns depending on the green light cases, laser sensors and servos for the collection of victims.



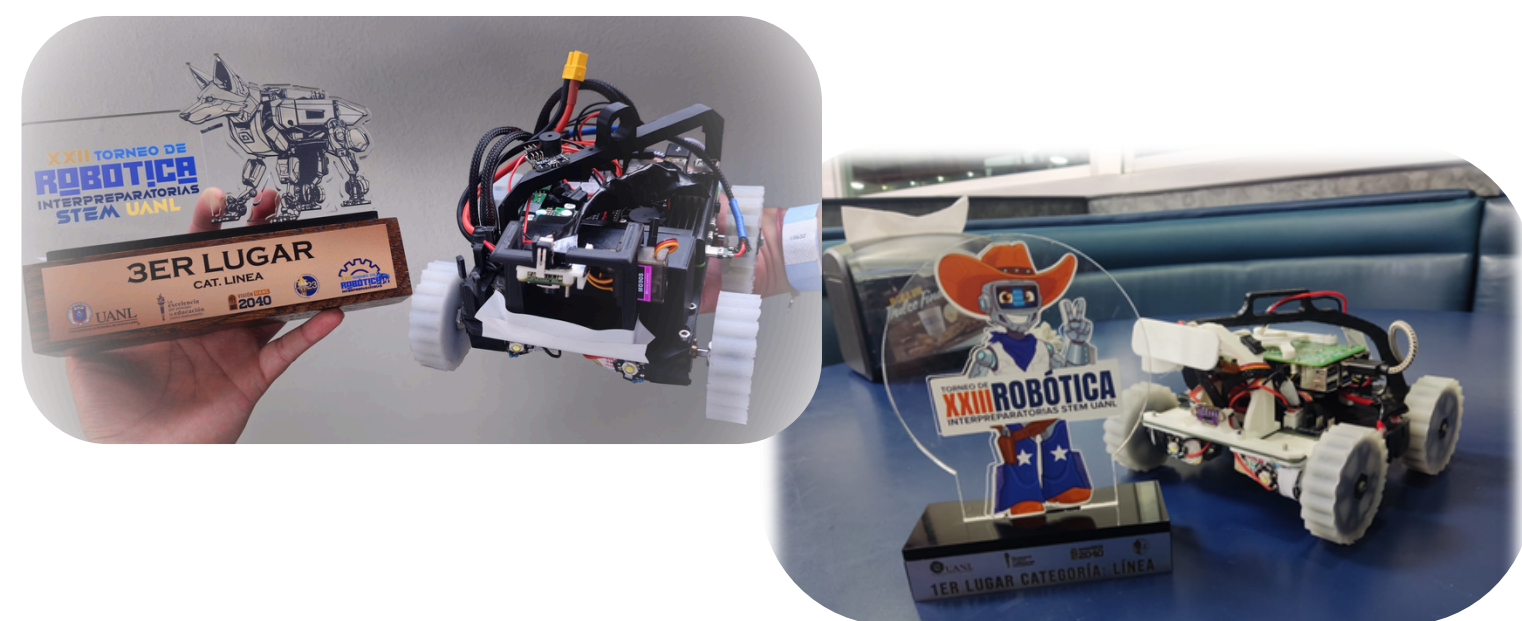
Preparatoria 20 UANL
Coordinador: Adolio Sanchez Villarreal



ALPHA.RLINE

Experiencias Previas

In our initial participation, we obtained 3rd place. Although the performance was competitive, we managed to gain experience and carried out a post-competition analysis, realizing that we had a very poor center of mass in addition to finding weaknesses in the software, problems that are not noticeable at first glance but that failed on the track. For our second version, we optimized both the traction and weight distribution of our robot, completely changing the main power supply method and patching various problems at the software level.



alpha¹ RESCUE LINE

D36V50F12.
12V regulator, supplies DC power for the other components.



D30V30F3.
3V to 3.7A regulator, its only job is to keep 2 LEDs lit to illuminate the track.



D36V50F5.
5V to 5A regulator, only provides power to the raspberry pi5.



x4 MOLICEL P28A.
18650 type cells are used in series to provide the voltage required by the robot.



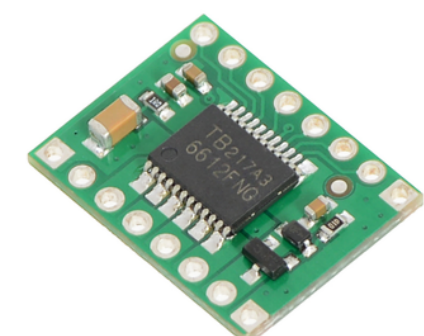
Rpicam Model3 Wide.
It handles the entire view of the track and is used in conjunction with the Raspberry Pi 5.



Raspberry pi5.
Their job is to track the line using the camera and send data to the teensy.



x2 TB6612FNG.
Used for motor control, regulating speed and direction.



x2 LED Smd Blanco 3w.
It helps considerably in the lighting of the track, this avoids irregularities in the perceived data.



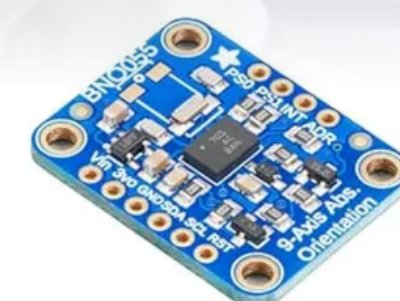
USB Acelerador.

Este dispositivo TPU nos permite correr modelos de IA desarrollados por nosotros mismo.



BNO055.

Este giroscopio ayuda a que los giros en zona de evacuación sean precisos



Switch ON/OFF.

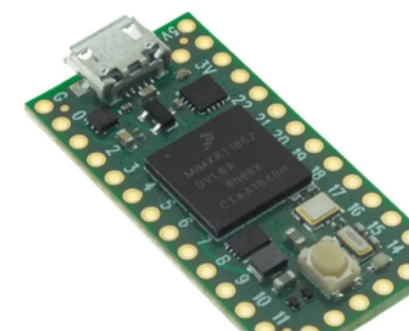
Nos permite pausar y reactivar el robot físicamente sin necesidad de detener el programa a nivel software



x4 Pololu 20DX46L.
Motors, the robot's main actuator, cause it to navigate on the track.



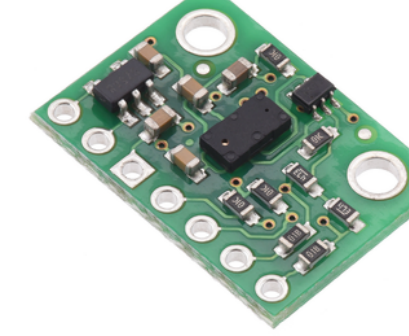
Teensy 4.0
Controls the robot's actuators and sensors, motors, laser sensor, servo, etc.



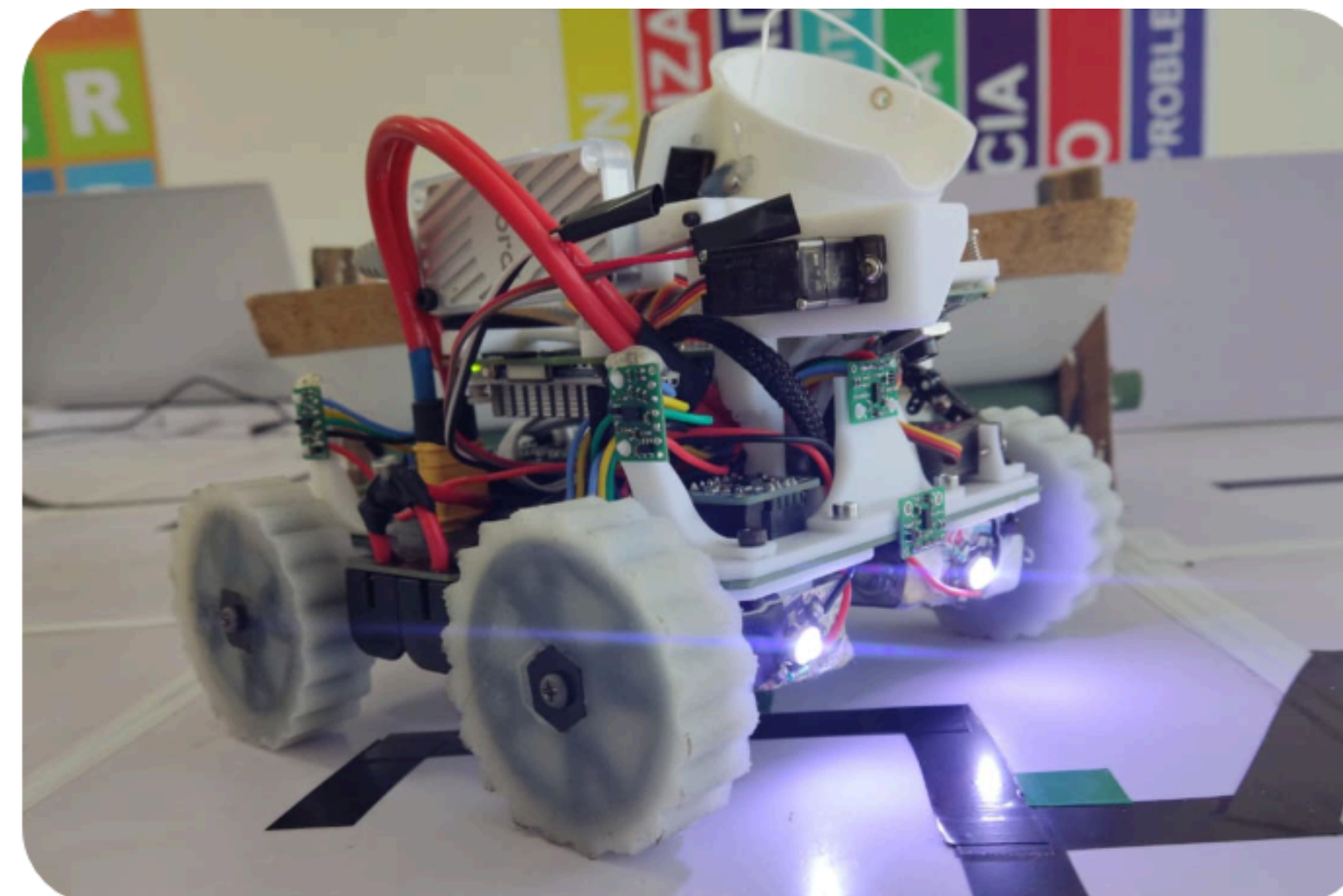
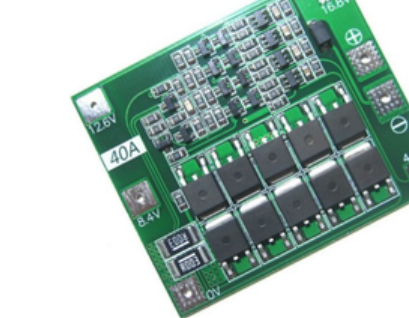
x3 Servo MG90S.
It helps control the camera's tilt angle to adjust for an ideal view, and is also used in the victim collection system.



x4 VL53L4CD.
These sensors help us detect obstacles and follow walls to exit the evacuation zone.

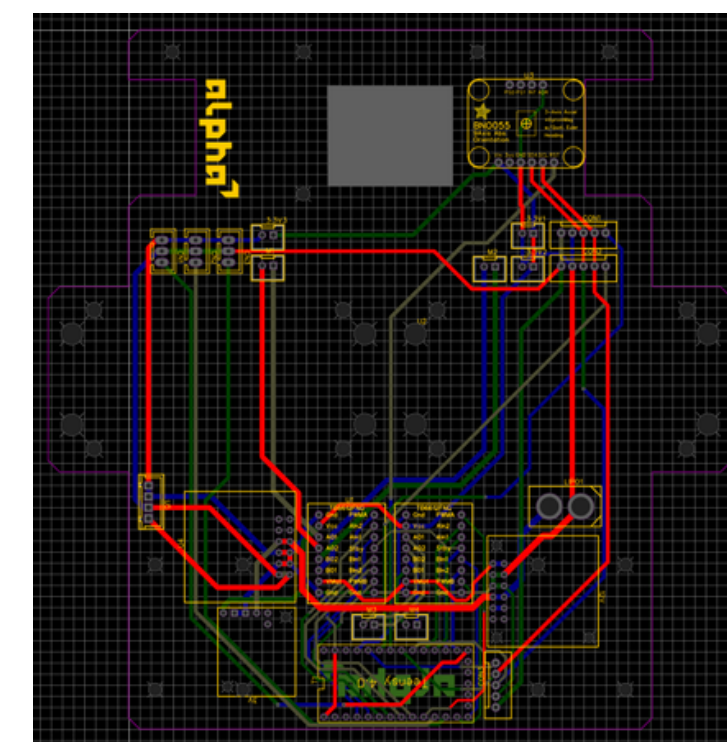
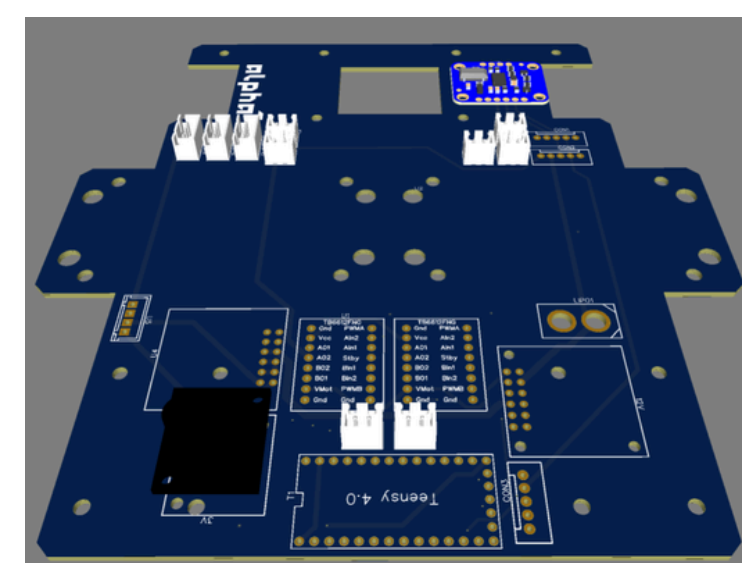


BMS 4S 40A.
It helps us to monitor and protect the cells against discharges and overloads.



Innovación

Our main mechanical innovation was the design of our custom PCB, as we had to develop it in tandem with the mechanical design. This allowed us to achieve shorter physical connections and, consequently, faster communication between components. In addition to fulfilling its electronic function satisfactorily, its role as the main chassis is crucial, making the robot more stable and secure. Another benefit is the easy access to all the robot's connections, which allows for faster external modifications.



SOFTWARE

The robot uses two languages: Python on the Raspberry Pi, responsible for image processing and decision-making, and C++ on the Teensy, responsible for motor control.

The Raspberry Pi captures frames from the camera and processes them to detect the black line, green turns, and red signals, as well as to control AI models running on a USB Coral Accelerator. Based on its detections, it calculates a direction value and sends it to the Teensy via serial communication. The Teensy receives this value and converts it into motor speeds, while also independently reading its own distance sensor to react to obstacles. Both controllers run in a continuous loop throughout the entire process.

Diagrama de Flujo - Line Follower:

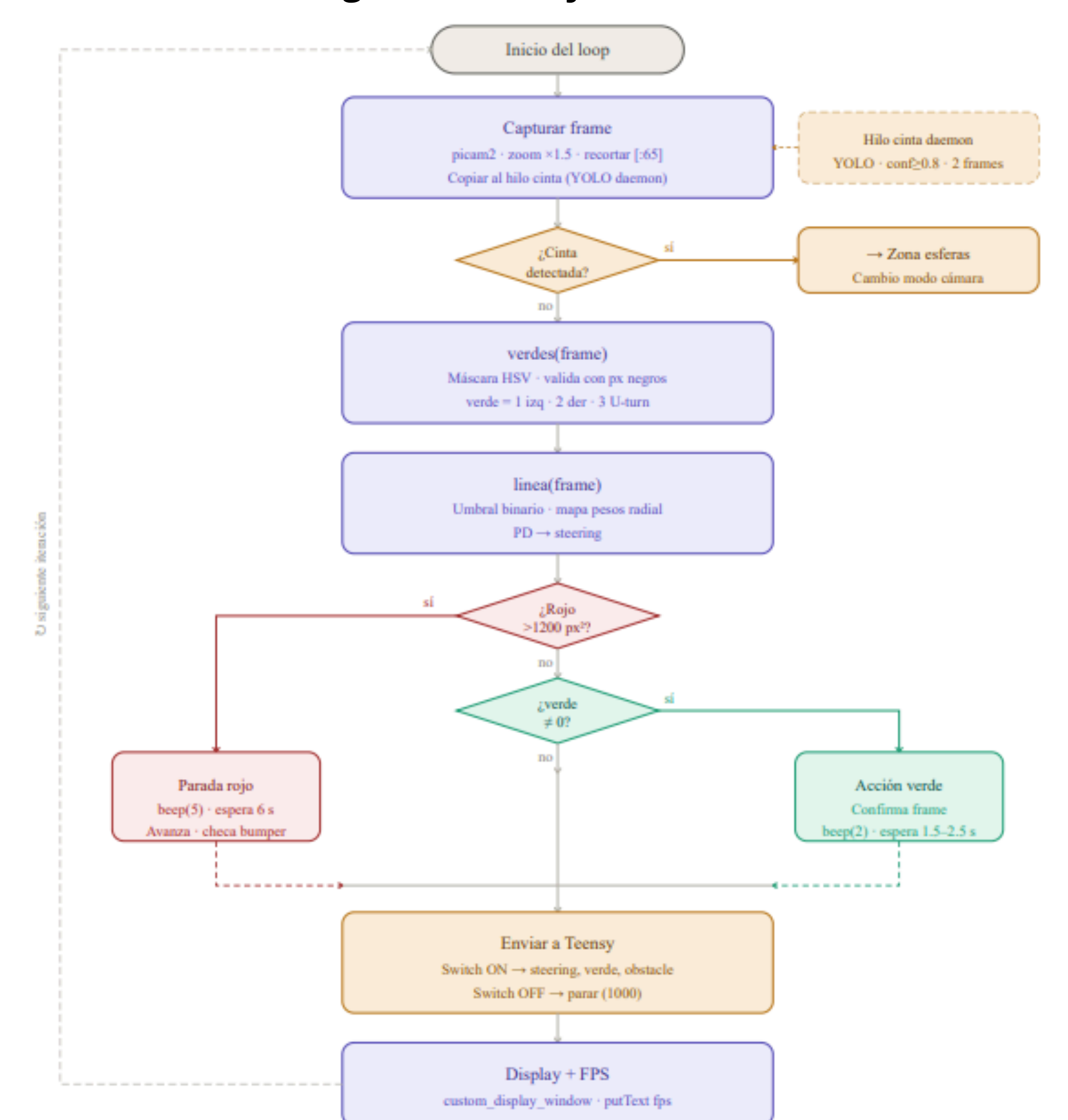


Diagrama de Flujo - Evacuation Zone:

